DATA VISUALIZATION AND DEEP LEARNING PROJECT

Predicting the Maturity Level of Netflix Titles from Metadata and Plot Descriptions

An end-to-end study: EDA, 7-stage visualization and a Multi-Layer Perceptron on the Netflix Movies and TV Shows dataset

Submitted by	Sudharshan N
Roll number	24AD119
Programme	B.Tech Artificial Intelligence and Data Science, III Year
Institution	KPR Institute of Engineering and Technology
Dataset	netflix_titles.csv (8,807 titles, 12 columns)
Model	Multi-Layer Perceptron (feed-forward neural network)

Executive Summary

This project studies the public Netflix catalogue (8,807 movies and TV shows) and builds a deep learning model that predicts a title's **maturity group** (Kids, Teens or Adults, derived from its official rating) using only metadata and the plot description, never the rating itself. After cleaning, 8,717 titles were used, split 70/15/15 into train, validation and test sets. A Multi-Layer Perceptron trained on 155 engineered features reached **70.3% test accuracy, macro-F1 of 0.731 and macro ROC-AUC of 0.856**, ahead of logistic regression, a random forest and a majority-class baseline. Kids content is identified well (AUC 0.962); separating Teens from Adults is the hard part, accounting for 88% of all errors.

1. Step 1: Dataset Selection

Source: the 'Netflix Movies and TV Shows' dataset, published on Kaggle by S. Bansal [1]. It is a mixed dataset: numerical (release_year, duration), categorical (type, country, rating), multi-label (listed_in genres, cast, director) and free text (description), so it supports both tabular and text analysis.

Feature requirement (15+): the raw file has 12 columns, so to meet the 15-feature requirement it was expanded with 17 engineered fields (see below), giving 29 data fields before encoding and **155 model input features** after encoding. This is stated openly because the raw file alone is below 15 columns.

Raw column	Type	Use in this project
show_id, title	ID / text	Identifier; title length engineered
type	Categorical	Movie vs TV Show flag
director, cast, country	Text lists	Counts, presence flags, primary country (one-hot)
date_added	Date string	Parsed into year_added, month_added, lag_years
release_year	Numeric	Used directly
rating	Categorical	**Target source** (grouped into 3 classes); not used as an input
duration	Text ('90 min', '2 Seasons')	Split into movie_minutes and n_seasons
listed_in	Multi-label text	Top-25 genres, multi-hot; n_genres
description	Free text	Length, VADER sentiment, TF-IDF + SVD (100 dims)

Engineered fields (17): is_movie, lag_years, year_added, month_added, movie_minutes, n_seasons, n_genres, n_cast, n_countries, n_directors, has_director, has_cast, primary_country, desc_words, desc_chars, title_len, sentiment.

Problem statement and target. Ratings are grouped as **Kids** (TV-Y, TV-Y7, TV-Y7-FV, TV-G, G), **Teens** (TV-PG, PG, TV-14, PG-13) and **Adults** (TV-MA, R, NC-17). Titles marked NR/UR or with no rating (90 rows) were removed as their maturity is unknown. Genre tags such as 'Kids' TV' are legitimate catalogue metadata and are not derived from the rating, so there is no target leakage. The task is therefore a 3-class classification problem.

2. Step 2: Data Understanding and Preprocessing

Exploratory data analysis started with the structure of the data (8,807 x 12), data types, missing-value profile and class balance (Figures 1 to 3 below). The cleaning steps and their effect are summarised here.

Issue	Finding	Treatment
Missing values	director 2634 (29.9%), cast 825 (9.4%), country 831 (9.4%), date_added 10, rating 4, duration 3	Text lists filled with 'Unknown' with has_director / has_cast flags kept; missing date_added falls back to release_year; month to the mode
Shifted rows	3 rows had the duration typed into the rating column	Moved to duration; their rating set to missing
Duplicates	0 exact, 0 on (title, type, release_year)	None found; check kept in pipeline
Unrated titles	90 titles (NR, UR or missing)	Dropped (target unknown); 8,717 rows remain

Issue	Finding	Treatment
Outliers (IQR rule)	movie_minutes (movies only): 388; n_seasons (TV only): 258; n_cast: 122; lag_years: 962; desc_words: 64	Winsorised at the 1st/99th percentile, bounds learned on the training set only
Transformation	Dates, durations and list fields stored as text	Parsed to numbers; sentiment via VADER [6]

Encoding and scaling. The binary flags stay 0/1; the 14 numeric features are standardised (z-score) with a scaler fitted on the training set only; the primary country is one-hot encoded (top 12 plus 'Other' = 13 columns); genres are multi-hot encoded (top 25); descriptions are converted by TF-IDF (1 to 2-grams, 8,000 terms) [7] and compressed with truncated SVD (latent semantic analysis [11]) into 100 dimensions that retain 12.6% of the TF-IDF variance. Total: 155 features.

Splitting. A stratified 70/15/15 split with a fixed seed keeps the class proportions identical in every subset. All transformers were fitted on the training portion and then applied to validation and test data to prevent leakage.

Subset	Rows	Kids	Teens	Adults
Train	6101	635	2660	2806
Validation	1308	136	570	602
Test	1308	137	570	601

3. Step 3: Data Visualization

The brief requires all 7 stages of visualization. This project follows Ben Fry's seven-stage process [2]; the table maps each stage to concrete work. Fourteen static figures and three interactive Plotly dashboards were produced (more than the 5 required).

Stage	What was done	Where
1. Acquire	Loaded netflix_titles.csv; profiled missing values	Fig. 1
2. Parse	Parsed dates, durations and list fields into numeric/categorical columns	01_preprocess.py, Fig. 2
3. Filter	Removed unrated titles; inspected and treated outliers	Fig. 6
4. Mine	Correlations, pairwise relations, genre-by-maturity statistics, text n-grams	Figs. 7 to 9, 11
5. Represent	Chosen chart type per data type: bars, histograms, boxplots, heatmaps, word clouds	Figs. 2 to 13
6. Refine	Redesigned a default chart: sorting, labels, colour coding	Fig. 14
7. Interact	Hover, zoom and drill-down dashboards in Plotly	figures/interactive_*.html

1. Missing values per column (Acquire)

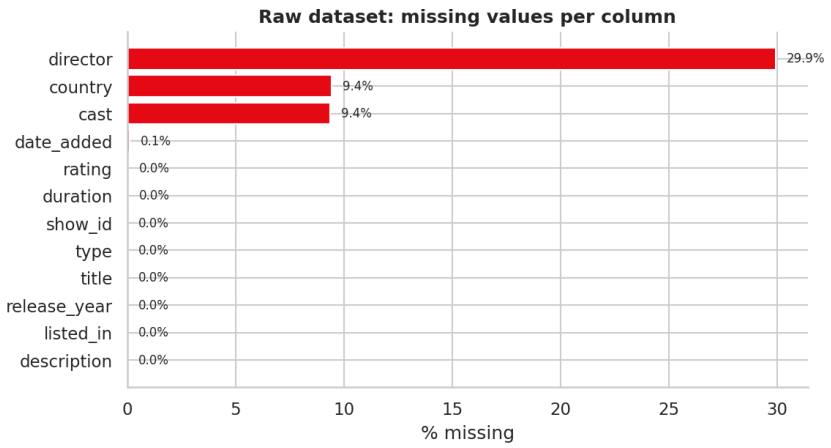


Figure 1: Missing values per column (Acquire)

What it shows: The percentage of empty cells in each raw column.

Why created: To decide what needs imputation before any modelling.

Insights: Director is missing in about 30% of titles and cast and country in about 9.4% each; every other column is almost complete. Because absence itself may carry information (documentaries and stand-up specials often have no cast), presence flags were kept alongside the 'Unknown' fill.

2. Content type and additions per year (Parse, Represent)

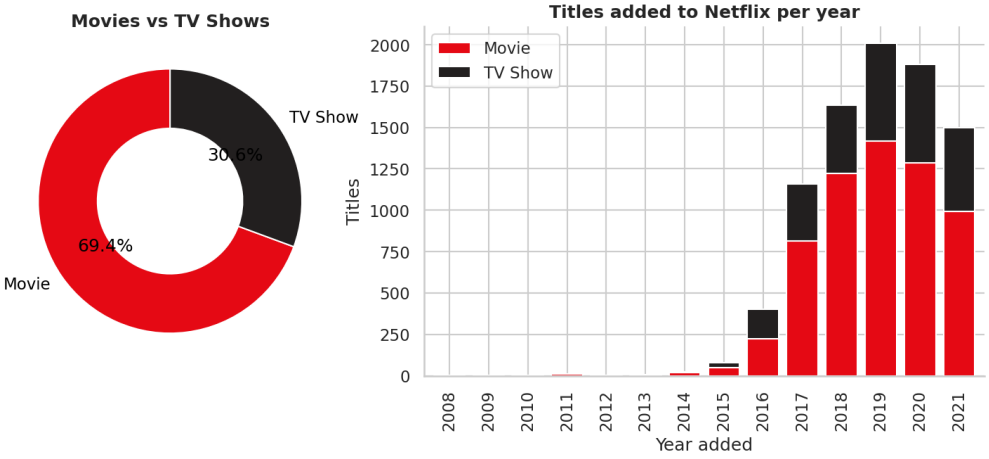


Figure 2: Content type and additions per year (Parse, Represent)

What it shows: A donut of Movies versus TV Shows and a stacked bar of titles added to Netflix each year.

Why created: To understand the catalogue composition and its growth over time.

Insights: Movies make up 69.4% of the catalogue and TV shows 30.6%. Additions rose steeply from 400 titles in 2016 to a peak of 2,010 in 2019, then fell slightly; the file is a 2021 snapshot, so 2021 is a partial year.

3. Rating distribution and target classes

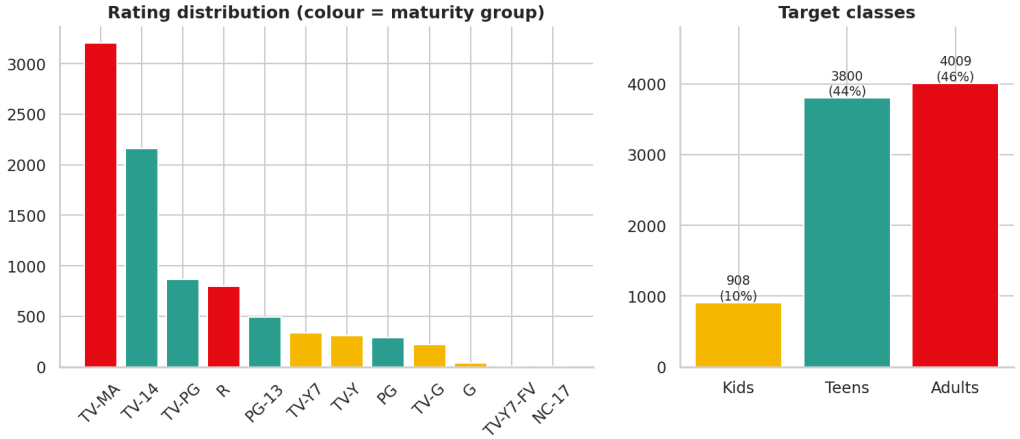


Figure 3: Rating distribution and target classes

What it shows: Counts of each rating, colour-coded by the maturity group, and the resulting class sizes.

Why created: To define the classification target and check for class imbalance.

Insights: TV-MA (3,207) and TV-14 (2,160) dominate. The target is Adults 46.0%, Teens 43.6%, Kids 10.4%: Kids is a clear minority, so accuracy alone would be misleading and macro-F1 is used as the main metric.

4. Top 10 producing countries

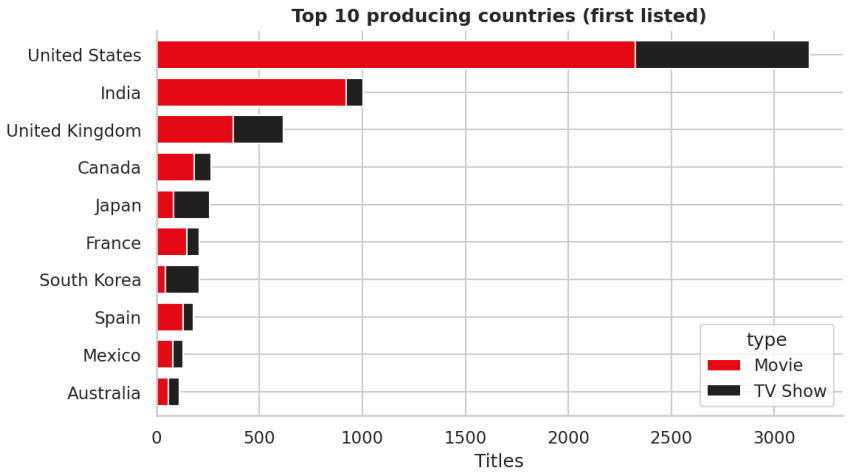


Figure 4: Top 10 producing countries

What it shows: Title counts by first-listed country, split by Movie and TV Show.

Why created: To see geographic coverage and decide how many countries to one-hot encode.

Insights: The United States leads with 3,171 titles, followed by India (1,004) and the United Kingdom (618); 830 titles have no country. The distribution has a long tail, so only the top 12 countries plus 'Other' were encoded.

5. Release year and release-to-addition lag

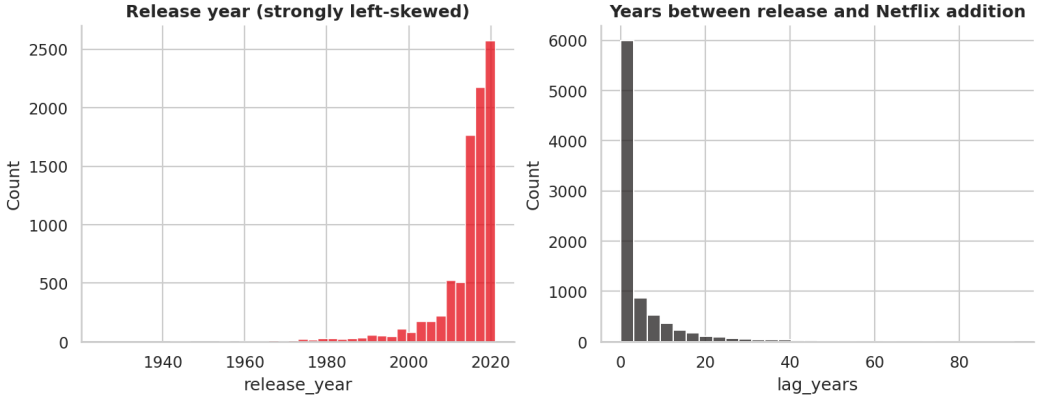


Figure 5: Release year and release-to-addition lag

What it shows: Histograms of release_year and of the years between release and arrival on Netflix.

Why created: To check skewness and the recency of the catalogue.

Insights: The catalogue is recent: median release year 2017 and 70.8% of titles released in 2015 or later. The median lag is one year and 55.2% of titles were added within a year of release, but a long tail of classic titles exists (back to 1925).

6. Outlier box plots by maturity group (Filter)

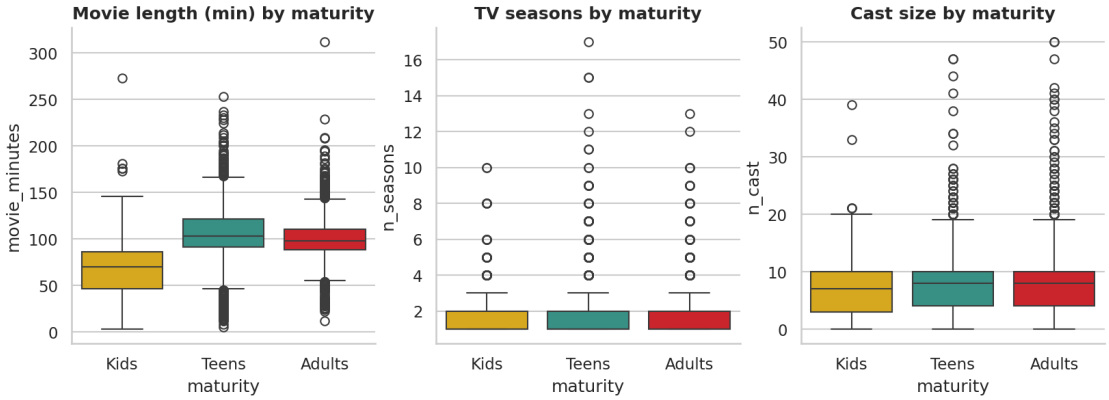


Figure 6: Outlier box plots by maturity group (Filter)

What it shows: Movie length, number of seasons and cast size by maturity group, with outliers as dots.

Why created: To find extreme values that could destabilise neural network training.

Insights: Median movie length is 69.5 min for Kids versus 103 (Teens) and 98 (Adults); Kids films are visibly shorter. Many extreme points exist (388 movie lengths, 258 season counts, 122 cast sizes by the IQR rule), which motivated winsorising at the 1st/99th percentile.

7. Correlation heatmap (Mine)

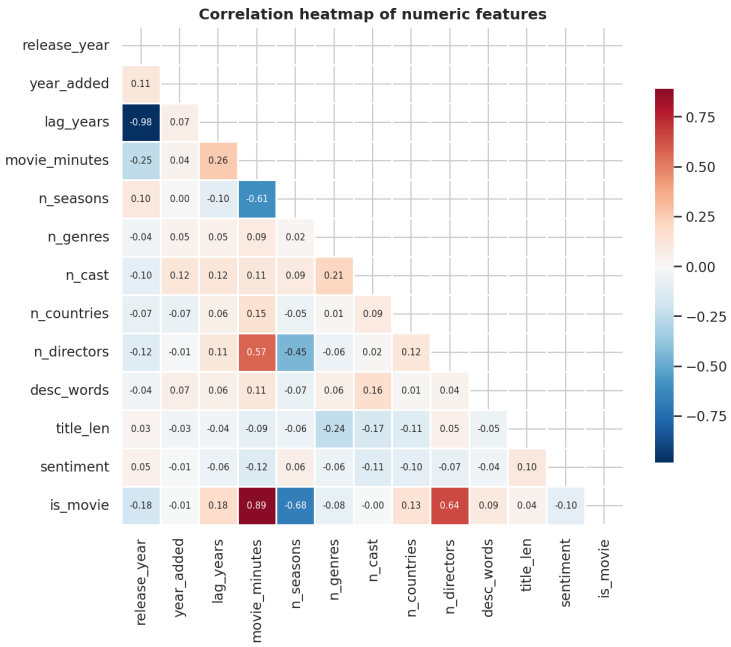


Figure 7: Correlation heatmap (Mine)

What it shows: Pairwise Pearson correlations between the numeric features.

Why created: To detect redundant features and linear relationships.

Insights: release_year and lag_years are almost perfectly negatively correlated (-0.98), and is_movie, n_seasons and movie_minutes are structurally linked (-0.68 and -0.61) because each applies to one content type only. Beyond these, correlations are weak, so little linear signal exists in the numeric block alone and a non-linear model is justified.

8. Pairplot of key numeric features (Mine)

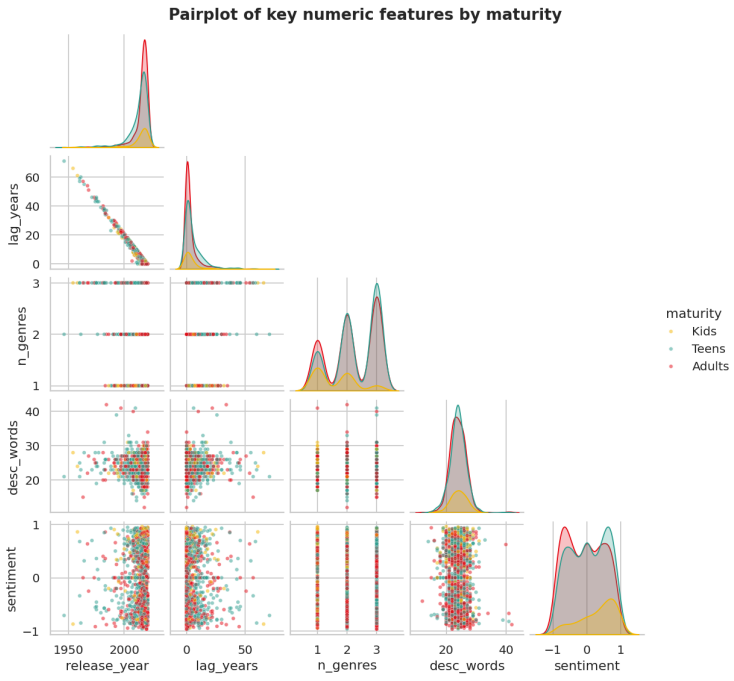


Figure 8: Pairplot of key numeric features (Mine)

What it shows: Scatter and distribution plots for six features, coloured by maturity group.

Why created: To see whether the classes separate in the numeric feature space.

Insights: The three classes overlap heavily in every pair. Only sentiment shows a systematic shift, with Kids descriptions leaning positive. Numeric metadata alone therefore cannot separate the classes; genre and text features are needed.

9. Genre by maturity heatmap (Mine)

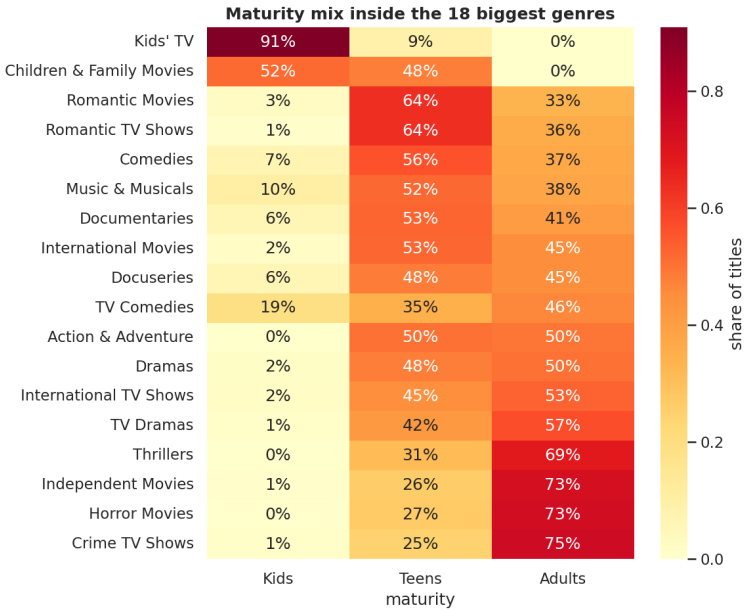


Figure 9: Genre by maturity heatmap (Mine)

What it shows: For the 18 largest genres, the share of titles falling in each maturity group.

Why created: To test which categorical tag is the strongest signal for the target.

Insights: 'Kids' TV' is 91% Kids and 'Children and Family Movies' has no Adult titles at all, while Crime TV Shows (75%), Independent Movies (73%), Horror (73%) and Thrillers (69%) are mostly Adults. Genre is the strongest predictor, which the later feature-importance results confirm.

10. Word clouds of descriptions by maturity (Represent, text)

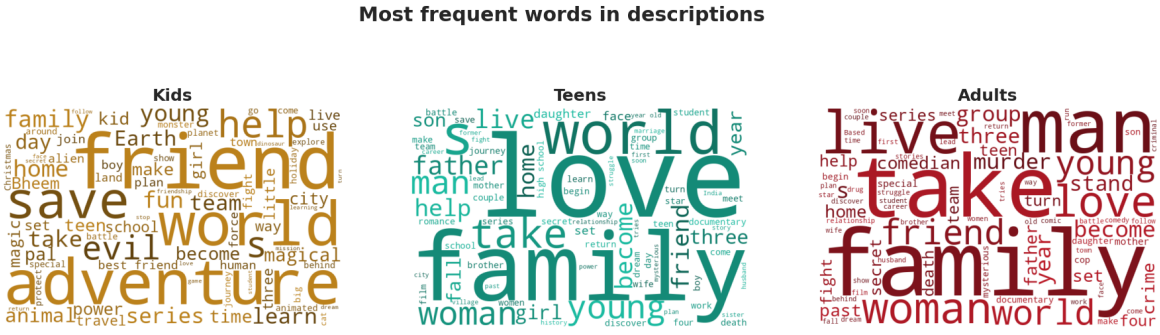


Figure 10: Word clouds of descriptions by maturity (Represent, text)

What it shows: The most frequent words in the descriptions of each group.

Why created: To compare the vocabulary used by each audience group.

Insights: Kids descriptions stress adventure, friend, save, magic, fun and animal; Adults descriptions contain murder, crime, cop, comedian and drug. Words such as family, love and world are common to all groups, so single words are only a partial signal.

11. Top bigrams by maturity group (Mine, text)

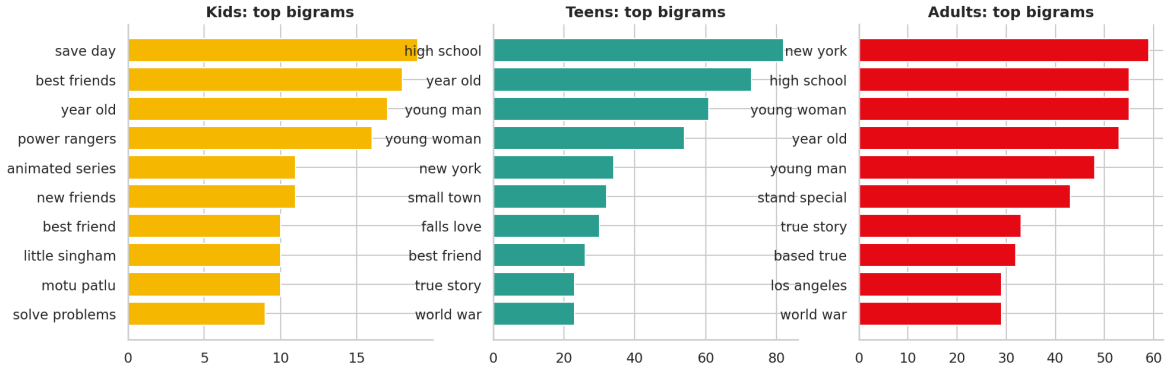


Figure 11: Top bigrams by maturity group (Mine, text)

What it shows: The ten most frequent two-word phrases per group.

Why created: N-grams capture phrases that word clouds miss.

Insights: Kids: 'save day', 'best friends' and 'power rangers'. 'High school' is frequent in both Teens (82) and Adults (55), a lexical overlap that explains much of the Teens-versus-Adults confusion in the model.

12. Sentiment and description length

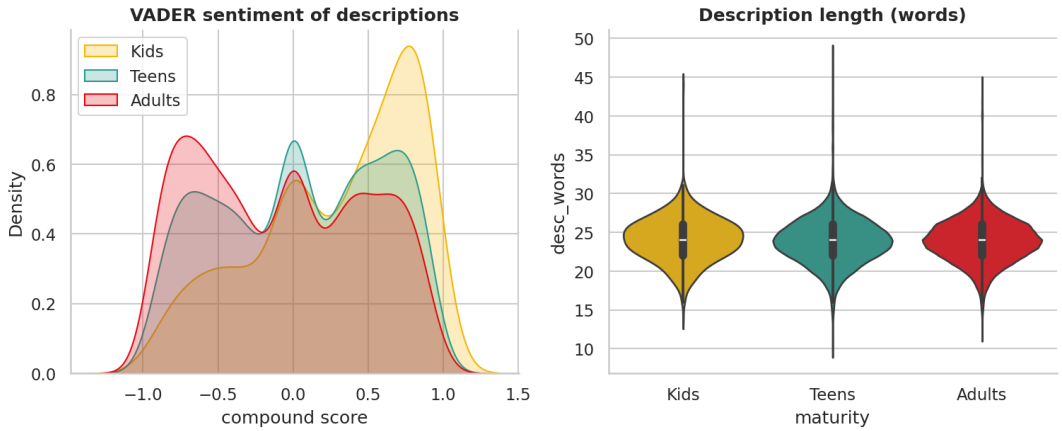


Figure 12: Sentiment and description length

What it shows: VADER compound sentiment distribution (left) and description length in words (right) per group.

Why created: To test whether tone or verbosity differs by audience.

Insights: Mean sentiment falls steadily with maturity: +0.28 for Kids, +0.05 for Teens and -0.08 for Adults. Description length is almost identical across groups (about 24 words), so it carries no signal.

13. Maturity mix of Movies versus TV Shows

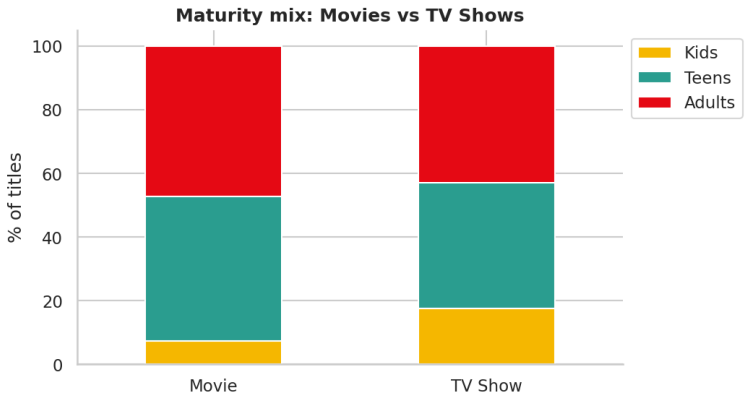


Figure 13: Maturity mix of Movies versus TV Shows

What it shows: The share of each maturity group inside the two content types.

Why created: To check whether content type is related to the target.

Insights: Kids content is a much larger share of TV shows (17.5%) than of movies (7.3%), while Teens and Adults are roughly balanced in both. Content type is therefore a useful but modest feature.

14. Refine: default versus redesigned chart

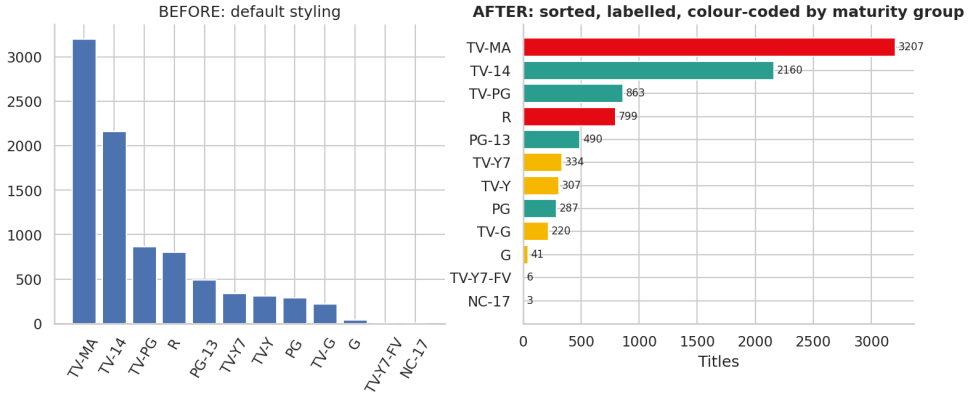


Figure 14: Refine: default versus redesigned chart

What it shows: The same rating counts drawn with default styling (left) and after refinement (right).

Why created: To demonstrate the Refine stage: sorting, direct labels and meaningful colour.

Insights: Sorting bars, using horizontal orientation for long labels, annotating counts and colouring by maturity group make the chart readable at a glance without a legend.

15. Interactive dashboards (Interact)

Three Plotly HTML files in the figures folder let the viewer hover, zoom and filter: **interactive_sunburst.html** (click to drill from type to maturity group to rating), **interactive_timeline.html** (additions per year by maturity group) and **interactive_scatter.html** (hover to see the title, rating and genres of each point). They open in any web browser and complete the Interact stage.

4. Step 4: Model Building

Model choice. A Multi-Layer Perceptron (MLP) is a feed-forward deep neural network [3] suited to the mixed tabular and text-embedding input. The final network has an input layer of 155 features, hidden layer(s) of size (64,) with ReLU activation, and a 3-unit softmax output; it has 10,179 trainable parameters. It is trained with the Adam optimiser [4] and cross-entropy loss, with a mini-batch size of 64. The model was implemented with scikit-learn [5]; the network is trained one epoch at a time so that train and validation loss and accuracy can be recorded after each epoch, and training stops early when validation loss has not improved for 20 epochs, restoring the best weights.

Feature analysis. Features fall into five groups: 14 numeric, 3 binary flags, 13 country indicators, 25 genre indicators and 100 text components. The EDA showed that numeric metadata is weak (Figures 7 and 8), genre is strong (Figure 9) and text and sentiment add complementary tone (Figures 10 to 12).

Hyperparameter tuning (basic). A grid search of 24 configurations was run over hidden-layer structure ((32), (64), (64, 32), (128, 64)), L2 penalty alpha (0.0001, 0.01, 0.1) and learning rate (0.0003, 0.001), each trained with early stopping and ranked by **validation macro-F1**. Validation scores ranged from 0.697 to 0.721. The widest network (128, 64) reached its best validation loss after only 5 to 15 epochs and scored lower on average (macro-F1 0.708) than the single 64-unit layer (0.718), so a small network with stronger regularisation won.

Rank	Hidden layers	alpha	Learning rate	Best epoch	Val loss	Val acc	Val macro-F1
1	(64,)	0.1	0.0003	58	0.6278	0.702	0.721
2	(32,)	0.0001	0.0003	43	0.6315	0.701	0.721
3	(64,)	0.0001	0.001	10	0.6311	0.704	0.720
4	(64,)	0.01	0.001	10	0.6296	0.704	0.719
5	(64,)	0.01	0.0003	31	0.6290	0.699	0.719

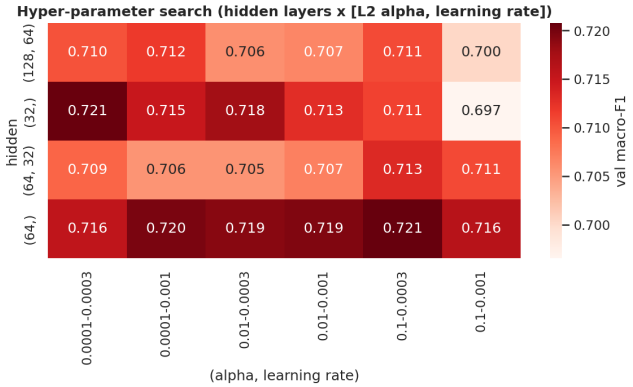


Figure 16: Hyperparameter search heat map (validation macro-F1)

Selected configuration: hidden layers (64,), alpha = 0.1, learning rate = 0.0003; best epoch 61 of 81 run.

Feature importance. Permutation importance was measured on the validation set: each feature group was shuffled 8 times and the drop in macro-F1 recorded. Text components were permuted together as one group, and country indicators as one group. A Random Forest [10] was fitted for comparison.

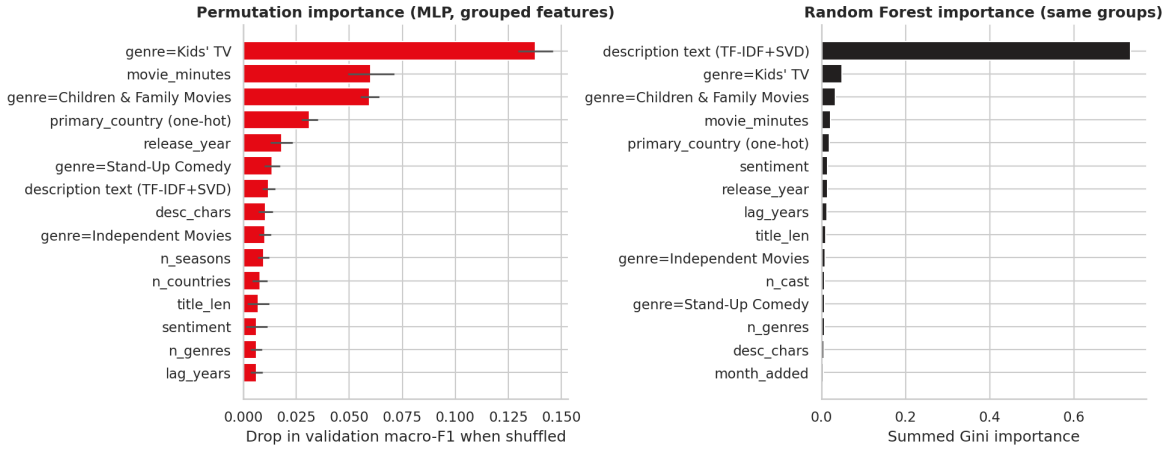


Figure 17: Feature importance, MLP permutation (left) and Random Forest (right)

Reading the chart. The 'Kids' TV' genre flag is by far the most important feature (macro-F1 falls by 0.138 when shuffled), followed by movie length (0.060), the 'Children and Family Movies' flag (0.060), the primary country (0.031) and release year (0.018). The Random Forest ranks the 100-dimensional text group first with a summed Gini importance of 0.74; this is largely an artefact, because impurity importance is biased towards continuous features with many candidate split points [9] and the sum is over 100 columns. The permutation view is the more trustworthy one here.

5. Step 5: Model Evaluation Visualization

All numbers below are on the untouched test set of 1,308 titles, unless stated otherwise. Four evaluation charts were required; six are provided (the error-distribution chart is for regression only, so a confidence distribution is used for error analysis instead).

18. Loss versus epoch

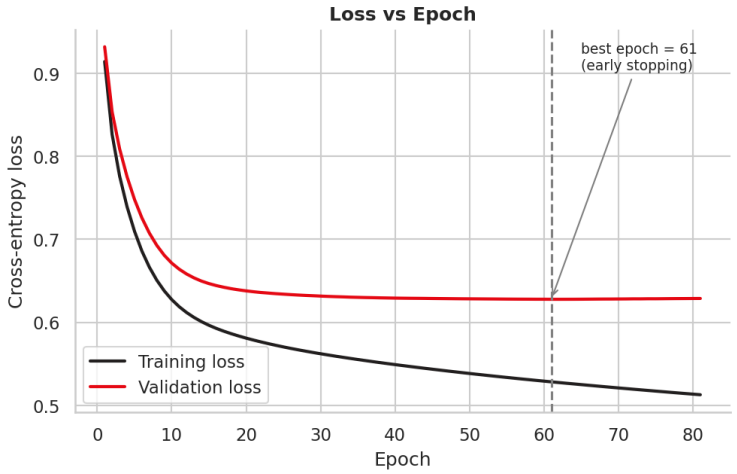


Figure 18: Loss versus epoch

What it shows: Training loss falls from about 0.92 to 0.51 while validation loss flattens near 0.628 and bottoms at epoch 61.

What it conveys: The widening gap shows the usual overfitting pattern. Early stopping at epoch 61 keeps the weights with the lowest validation loss and avoids memorising the training set.

19. Accuracy versus epoch

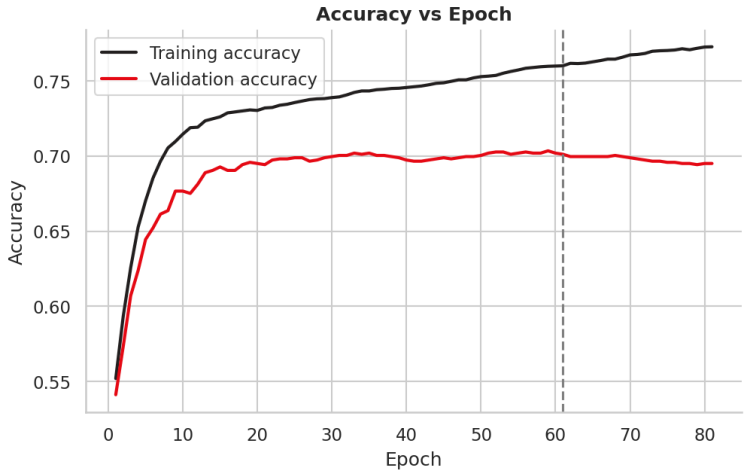


Figure 19: Accuracy versus epoch

What it shows: At the selected epoch training accuracy is 76.0% and validation accuracy 70.1%.

What it conveys: A gap of about 6 points confirms mild overfitting, but validation accuracy is stable rather than collapsing, so the model generalises reasonably well.

20. Confusion matrix

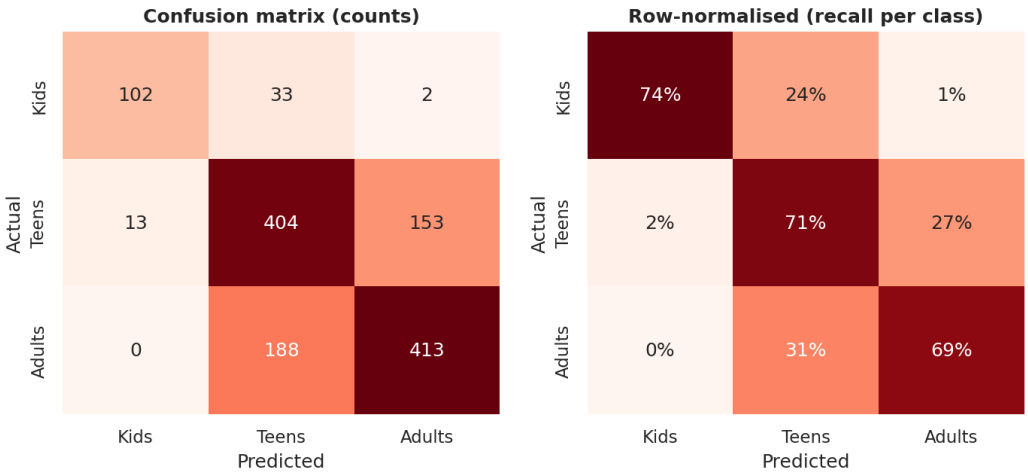


Figure 20: Confusion matrix

What it shows: Counts and per-class recall. Kids recall is 74.5% (102 of 137), Teens 70.9%, Adults 68.7%.

What it conveys: The model almost never confuses Kids with Adults (2 and 0 cases). Most errors, 341 of 389, are between Teens and Adults, which are adjacent maturity levels and share genres (drama, comedy) and vocabulary ('high school').

21. ROC curves and AUC (one-vs-rest)

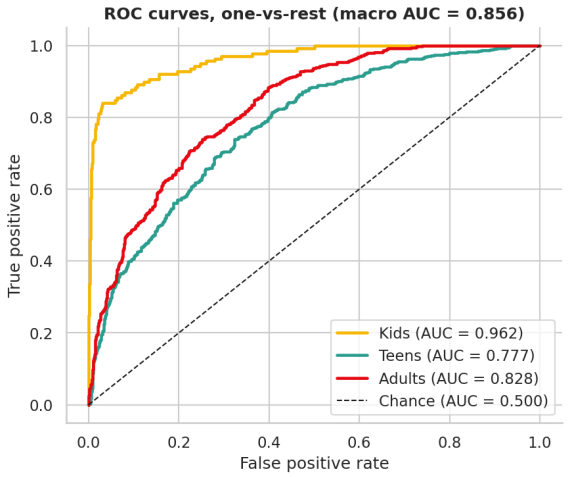


Figure 21: ROC curves and AUC (one-vs-rest)

What it shows: AUC is 0.962 for Kids, 0.828 for Adults and 0.777 for Teens; macro AUC is 0.856.

What it conveys: Kids is separated almost perfectly from the rest. Teens is the hardest class because it sits between the other two, which matches the confusion matrix. All curves lie far above the chance diagonal (AUC 0.5).

22. Model comparison

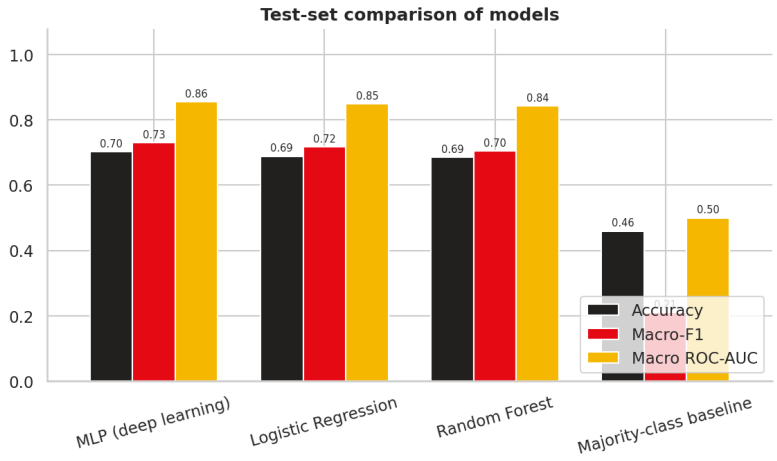


Figure 22: Model comparison

What it shows: Accuracy, macro-F1 and macro ROC-AUC for the MLP, logistic regression, random forest and a majority-class baseline.

What it conveys: The MLP is best on all three metrics (0.703 / 0.731 / 0.856) and far above the majority baseline (0.460 accuracy). Its lead over logistic regression is only about 1.4 accuracy points, which is within the sampling noise of a 1,308-row test set (roughly 1.3 points), so it should be read as 'at least as good', not as decisive superiority.

23. Error analysis: confidence of correct versus wrong predictions

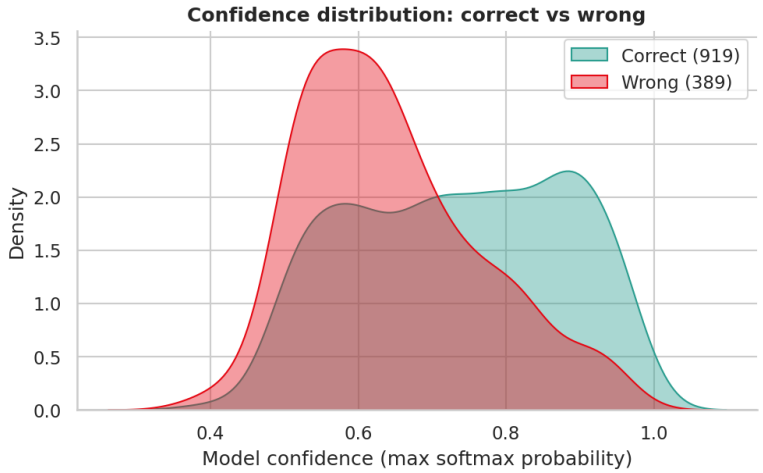


Figure 23: Error analysis: confidence of correct versus wrong predictions

What it shows: The distribution of the model's top predicted probability for correct and incorrect predictions. Mean confidence is 0.74 for correct and 0.64 for wrong predictions.

What it conveys: Wrong predictions are made with lower confidence, so low-confidence predictions could be routed for manual review in a real system. The overlap, however, shows the probabilities are only moderately informative.

Per-class test metrics

Class	Precision	Recall	F1-score	Support
Kids	0.89	0.74	0.81	137
Teens	0.65	0.71	0.68	570
Adults	0.73	0.69	0.71	601

Model	Accuracy	Macro-F1	Weighted-F1	Log loss	Macro AUC
MLP (deep learning)	0.7026	0.7308	0.7041	0.6140	0.8555
Logistic Regression	0.6888	0.7180	0.6902	0.6262	0.8494

Model	Accuracy	Macro-F1	Weighted-F1	Log loss	Macro AUC
Random Forest	0.6858	0.7047	0.6869	0.6886	0.8423
Majority-class baseline	0.4595	0.2099	0.2893	-	0.5000

6. Step 6: Conclusion and Future Scope

What was learned from the dataset

- The catalogue is movie-heavy (69.4%), recent (median release 2017) and dominated by US and Indian content; additions peaked in 2019.
- The target is imbalanced (Kids only 10.4%), so macro-F1 and per-class AUC are more honest than accuracy.

Key patterns and findings

- Genre is the dominant signal: 'Kids' TV' is 91% Kids, whereas Crime, Horror and Independent titles are 73 to 75% Adults.
- Movie length is a secondary signal (Kids median 69.5 minutes versus about 100 for others), and sentiment of the description falls steadily from Kids (+0.28) to Adults (-0.08).
- Numeric metadata is weakly correlated with maturity and two features are nearly redundant (release_year and lag_years, r = -0.98).

How the model performed

The MLP achieved 70.3% accuracy, 0.731 macro-F1 and 0.856 macro AUC on unseen data, compared with 46.0% for always predicting the majority class. It is slightly ahead of logistic regression and the random forest, but the margin is within test-set noise, which suggests the ceiling is set by the information available in the features rather than by model capacity. Kids detection is strong (F1 0.81); Teens versus Adults is the weak point (F1 0.68 and 0.71).

Challenges faced

- Raw column count (12) was below the 15-feature requirement, solved by feature engineering.
- Dirty rows (duration text stored in the rating column), heavy missingness in director and cast, and the many-valued list fields needed custom parsing.
- Fast overfitting: the widest networks peaked after only 5 to 15 epochs, which favoured smaller architectures, L2 regularisation and early stopping.
- The rating boundary between Teens and Adults (TV-14 versus TV-MA, PG-13 versus R) is inherently fuzzy and is partly decided by content (violence, language) that a short description does not reveal.

Future scope and improvements

- Replace TF-IDF + SVD with pre-trained transformer embeddings (for example BERT or sentence-transformers) or fine-tune a transformer on descriptions to capture context beyond word counts.
- Use PyTorch or TensorFlow to add dropout, batch normalisation and learning-rate schedules, and tune with Bayesian optimisation or k-fold cross-validation for tighter confidence intervals.
- Add richer features: cast and director identity embeddings, IMDb or Rotten Tomatoes scores, runtime of episodes, parental-guide keywords, and trailer or poster images (a multimodal model).
- Treat the problem as ordinal classification (Kids < Teens < Adults) and use class-weighted or focal loss to improve minority-class recall.
- Explain individual predictions with SHAP or LIME and deploy the model as a small FastAPI service that suggests a maturity group for a new title.

7. Step 7: Literature Review and References

Visualization practice. Fry [2] frames visualization as a seven-stage pipeline of acquiring, parsing, filtering, mining, representing, refining and interacting with data, and this project is organised around exactly that sequence. Plots were produced with Matplotlib [8], whose 2D graphics model underlies Seaborn and the word-cloud rendering used here.

Text representation. Term-weighting schemes such as TF-IDF [7] remain a standard way to turn documents into vectors, and latent semantic analysis [11] shows that a truncated singular-value decomposition of the term-document matrix yields a compact semantic space. Both are combined here to feed descriptions into a neural network. For tone, VADER [6] provides a lexicon- and rule-based sentiment score that needs no training data and works well on short texts such as these one-sentence descriptions.

Deep learning and evaluation. The theory of feed-forward networks, regularisation and early stopping is covered in the standard textbook by Goodfellow et al. [3]; training uses the Adam optimiser [4] through scikit-learn [5]. Random forests [10] serve as a strong tabular baseline. Permutation importance is used rather than impurity-based importance because the latter is known to favour continuous and high-cardinality features [9].

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